ChewieNN: Robust Keypoint Regression through Physical Imaging Simulation and Deep Residual Learning

Asher Wallen
Department of Biomedical Engineering
Duke University
Durham, NC 27708
alw107@duke.edu

Abstract

This project presents the Chewie Neural Network (ChewieNN), a specialized deep learning framework for high-precision feline facial keypoint regression despite inherent blurring and noise of standard cell phone cameras. Utilizing a ResNet-152 backbone, I investigate the trade-off between frozen generic feature extraction and subject-specific fine-tuning. To address real-world optical challenges, a physical simulation layer was integrated to introduce Gaussian noise and blur, each as a tunable hyperparameter. Experimental results showed that while a frozen backbone provides an acceptable baseline MSE of 0.0193, fine-tuning the deep residual blocks (layers 3 and 4) with a custom dataset of 29 subject-specific annotations for a feline specimen named Chewie reduces the test error to 0.001005, a 94.8% improvement. This study confirms that deep layer plasticity combined with subject-specific domain adaptation is essential for robust coordinate regression in non-professional imaging conditions.

1 Introduction

Automated pet monitoring systems require precise localization of facial features to interpret behavior and health. However, home environments often suffer from poor lighting, sensor noise, and motion blur, which significantly degrade the performance of standard computer vision models.

In this work, I implement ChewieNN, a regression model based on the ResNet-152 architecture. The project aims to solve two primary problems:

- Robustness: Designing a compound physical layer within the network architecture to simulate environmental degradation, forcing the model to learn noise-invariant features.
- Specialization: Transitioning from a general cat keypoint detector (trained on the Kaggle CAT dataset) to a subject-specific specialist model using manual annotations of a domestic cat (Chewie).

I hypothesize that unfreezing the final stages of a deep residual network—specifically the bottleneck blocks where high-level semantic features are encoded—is necessary to capture the unique geometric proportions of a specific subject, especially when viewed through the dirty lens of a simulated physical layer.

2 Related Work

The foundation of this project is the ResNet-152 architecture, which introduced the concept of "skip connections" or identity shortcuts to mitigate the vanishing gradient problem in extremely deep networks [2]. While ResNet is traditionally used for classification, recent adaptations have shown its efficacy in regression tasks, such as human pose estimation and keypoint localization. Its default architecture is presented in 1. Modifications to this architecture for my specific intended tasks are discussed in the Methods section.

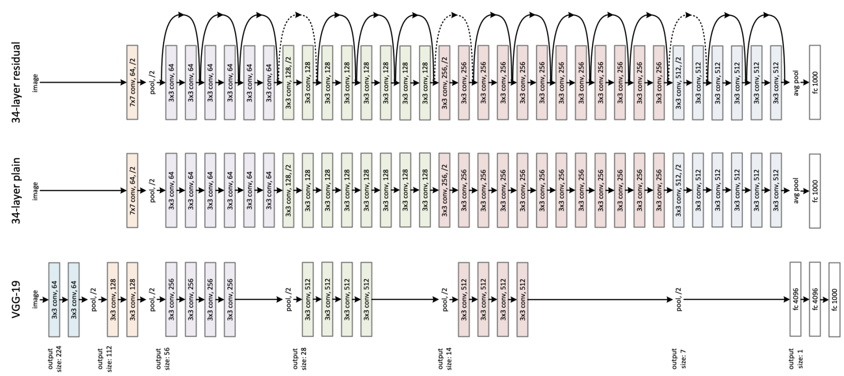

Figure 1: The standard ResNet-152 architecture as implemented by Microsoft [3]. The model features a 7x7 convolution followed by four stages of bottleneck residual blocks. In this default configuration, the network terminates in a Global Average Pooling layer and a 1,000-node fully-connected (fc) layer designed for ImageNet classification. For the purpose of the ChewieNN, this final fc layer is replaced with a custom linear regression head outputting 18 feline keypoint coordinates (x- and y-coordinates for 9 keypoints).

This study utilizes the Kaggle CAT dataset [1], a comprehensive collection of over 9,000 feline images annotated with facial landmarks. The dataset is accessed via the Kagglehub API, providing standard grounndtruths for feline facial regression. Previous work on this dataset has focused on general feline feature extraction, but this project seeks to extend that research by investigating models' ability to maintain reliable accuracy despite imperfect imaging techniques. It further seeks to fine-tune the regressive models in a subject-specific manner, a technique often used in facial recognition for personal security on human subjects but rarely applied to animals.

3 Methods

3.1 Dataset Preparation and Manual Annotation

The primary training data consisted of the Kaggle CAT dataset, which was split into training (60%), validation (20%), and test (20%) sets. To create the Chewie specialist model, I supplemented this with a custom dataset generated from high-resolution images of a specific feline subject: Chewie. Originally, I intended to compare two different subjects, but the image set for the second cat (Rey) was corrupted during transfer. Consequently, I shifted the focus of the fine-tuning trial exclusively to Chewie to ensure high data quality.

As shown in my data generation pipeline [4], 29 images were manually annotated using a custom coordinate system. An example is shown in 2, where the label is represented by the length-18 array

$$447673815510041312411484080508932111164$$

, matching the format of the .cat files present in the CAT dataset. Each image's .cat file contains 18 values representing the (x, y) coordinates for nine facial landmarks. These coordinates were normalized to the 224×224 input size required by the model.

Figure 2: Example of manual annotations of a Chewie image. Each ear (containing 3 points), each eye, and the mouth are represented by a unique color for visualization purposes.

3.2 Model Architecture and Training Loop

The ChewieNN architecture and its subsequent training trials were implemented in a modular Python framework [5]. The system consists of three distinct stages, which are expanded upon in further detail in 1:

- The physical layer A non-parametric stage applying Gaussian noise and a 3×3 blur kernel.
- The deep backbone: A ResNet-152 model [3] evaluated in Control (frozen), Unfrozen (generalist), and Fine-Tune (specialist) states.
- The regression head: A custom fully-connected linear layer replacing the standard classification head and forcing outputs into 1D vectors of size 18.

In the training loop, certain layers were frozen depending on the goal of the current phase of the ChewieNN (2). Specifically, the early bottleneck layers were always frozen while the deeper feature-associated layers remained frozen for the control phase but were unfrozen for the fine-tuning phases.

Table 1: Structural specifications of the ResNet-152 backbone sections [2].

Named Child	Description	Output Size (HxWxC)
conv1	7×7 Convolution	$112 \times 112 \times 64$
bn1	Batch Normalization	$112 \times 112 \times 64$
relu	Non-parametric Activation	$112 \times 112 \times 64$
maxpool	3×3 Max Pool	$56 \times 56 \times 64$
layer1	3 Residual Blocks	$56 \times 56 \times 256$
layer2	8 Residual Blocks	$28 \times 28 \times 512$
layer3	36 Residual Blocks	$14 \times 14 \times 1024$
layer4	3 Residual Blocks	$7 \times 7 \times 2048$
avgpool	Global Average Pool	$1 \times 1 \times 2048$

4 Results

The performance of the ChewieNN was evaluated across three experimental conditions to isolate the effects of model capacity and subject-specific data. Quantitative results were obtained by measuring

Table 2: Layer training status across experimental conditions.

Named Child	Control Status	Fine-Tuning Status
physical_layer	Constant	Constant
conv1, bn1	Frozen	Frozen
relu, maxpool	Non-parametric	Non-parametric
layer1, layer2	Frozen	Frozen
layer3, layer4	Frozen	**Trainable**
avgpool	Non-parametric	Non-parametric
fc	**Trainable**	**Trainable**

the Mean Squared Error (MSE) on an independent holdout test set containing a mixture of generic and subject-specific images.

4.1 Quantitative Analysis

As detailed in 3, the transition from a frozen backbone to an unfrozen state resulted in a massive reduction in error. The Fine-Tuned (Specialist) model achieved the highest precision, reaching an MSE of 0.001005.

Table 3: Comparative Test Loss (MSE) across experimental conditions.

Model Condition	Test MSE	Improvement
Control (Frozen Backbone)	0.019299	Baseline
Unfrozen (Generalist)	0.001070	94.46%
Fine-Tuned (Specialist)	**0.001005**	**94.79%**

4.2 Training Curve Convergence

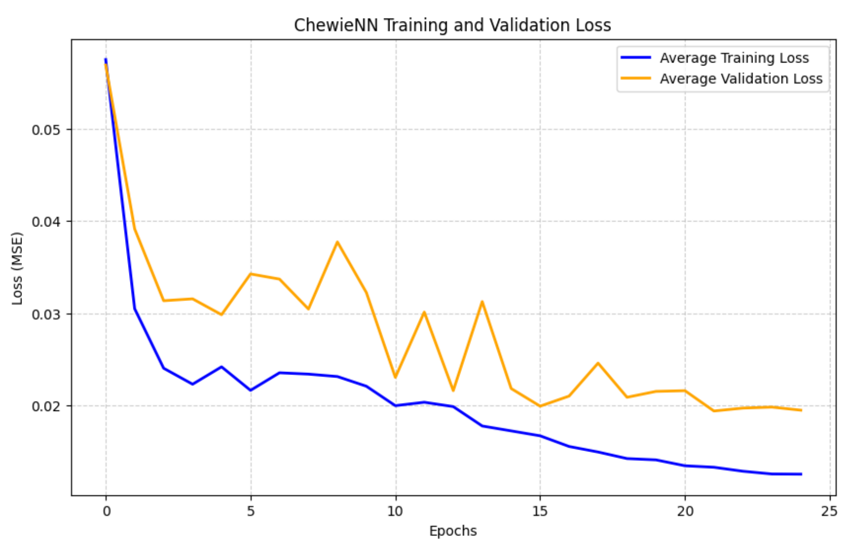

Figure 3: Training and validation loss for the Control Model (backbone frozen). While the model converges, it plateaus at a significantly higher MSE (≈ 0.019), indicating that a frozen generic feature extractor lacks the precision required for high-fidelity coordinate regression under optical degradation.

The training process for each model provided insight into the learning stability under simulated optical degradation.

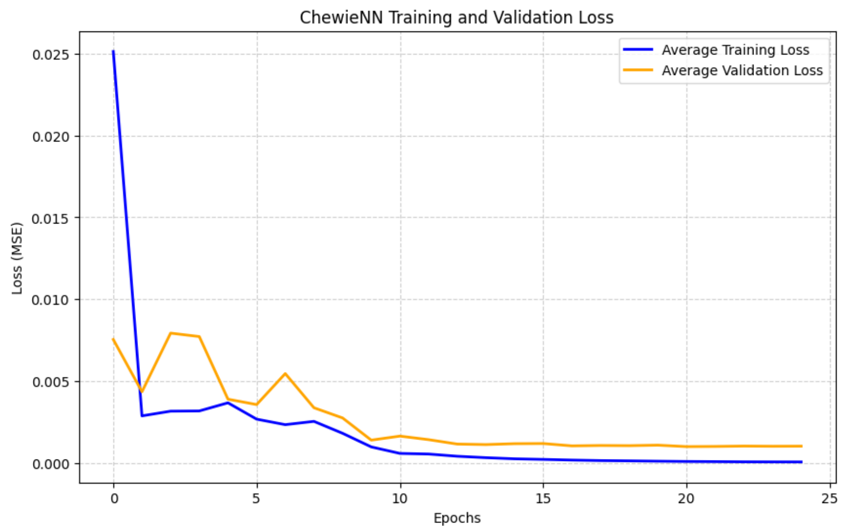

Figure 4: Training and validation loss for the Unfrozen Model (Generalist). By unfreezing residual layers 3 and 4, the model's representational capacity increases, leading to a rapid drop in MSE. This illustrates that deep semantic filters must remain plastic to effectively map feline geometry to keypoint coordinates.

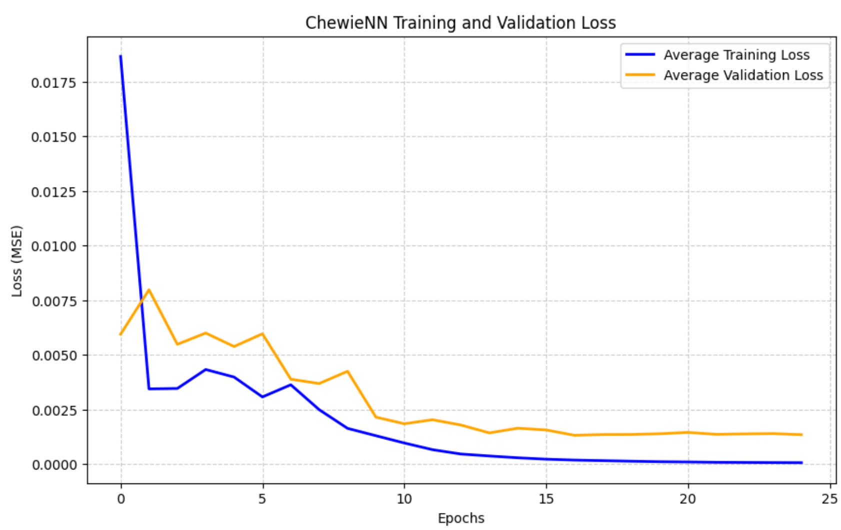

Figure 5: Training and validation loss for the fine-tuned Chewie specialist. The addition of subject-specific data alongside unfrozen residual blocks results in the lowest overall loss (≈ 0.001). The smoother convergence compared to the control trial suggests the model successfully adapted to the unique facial proportions of the test subject.

The Control Model (3) exhibited significant validation jitter and plateaued early, suggesting that the frozen ImageNet-derived filters were unable to resolve fine-grained feline landmarks through the physical layer's noise.

In contrast, the Unfrozen Model (4) showed a rapid descent in training loss, though it maintained slightly higher validation error than the specialist.

The Fine-Tuned Specialist (5) had by far the most stable convergence profile, successfully adapting its deep residual filters to the specific morphology of the subject while ignoring the stochastic perturbations of the Gaussian noise and blur.

5 Discussion

The results confirm that while ResNet-152 is an exceptional general-purpose feature extractor, the transition to high-precision subject-specific localization requires deep-layer plasticity.

The 94.79% reduction in error, which is visualized in 6, between the Control and Specialist models suggests that the fixed filters in the pre-trained ImageNet backbone are insufficient for the nuances of specific feline morphology. Additionally, the fact that the Specialist model slightly edged out the Unfrozen model—despite the subject-specific data making up less than 1% of the total training volume—highlights the power of domain-specific fine-tuning. By unfreezing layer3 and layer4, the network was able to resolve the specific geometry of unusually oriented "Chewie" images even through the simulated noise of the physical layer.

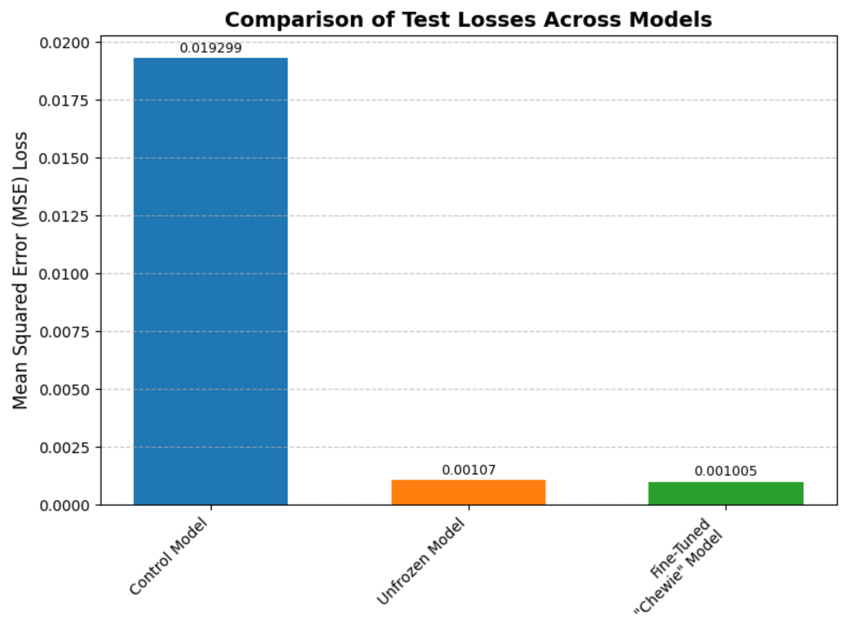

Figure 6: Comparative analysis of Mean Squared Error (MSE) on the independent holdout test set. The Specialist model achieves a 94.8% reduction in error relative to the Control, proving that subject-specific fine-tuning is the optimal strategy for precision home-monitoring applications.

6 Acknowledgements

Thanks to Dr. Roarke Horstmeyer and Josh Lerner for their support through this project. Thanks to the authors of the Kagglehub datasets that I used to train my models, and whose format I mimicked when creating my own annotated dataset. And most of all, thanks to Chewie for all the joy he brought me in his 11.5 years of life.

6.1 LLM Acknowledgement

Gemini (embedded in Google Colab) was used heavily in debugging and increasing the modularity of code in my Python notebooks, especially when transferring files between their Internet sources, my local devices, and Google Drive. The conceptual parts of each model were written by me, inspired by class materials as well as some further research into Pytorch architectures.

References

[1] Crawford, C., NAIN. (2018). *CAT Dataset*. Retrieved from Kaggle via Kagglehub. https://www.kaggle.com/datasets/crawford/cat-dataset/code

[2] He, K., Zhang, X., Ren, S. and Sun, J. (2016) Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, 27-30 June 2016, 770-778. https://doi.org/10.1109/CVPR.2016.90

[3] Microsoft. (2022). ResNet-152 Pre-trained Model. *Hugging Face Model Hub*. https://huggingface.co/microsoft/resnet-152

[4] Generation and Verification of Annotated Dataset. Python Notebook. https://github.com/AsherWallen/ChewieNN/blob/main/Generation_and_Verification_of_Annotated_Dataset.ipynb

[5] ChewieNN Code. Python Notebook. https://github.com/AsherWallen/ChewieNN/blob/main/Asher_Wallen_ChewieNN_Cod